CSC244 Project Phase 2
Text Based RAG/LLM State Selector Prototype

Phase 2:

Preliminary implementation using Google Colab.
Phase 2 will feature several handcrafted scenarios primarily building an environment to make it possible to evaluate latency and accuracy of several LLM models on Colab's T4 GPU.

TOOLS

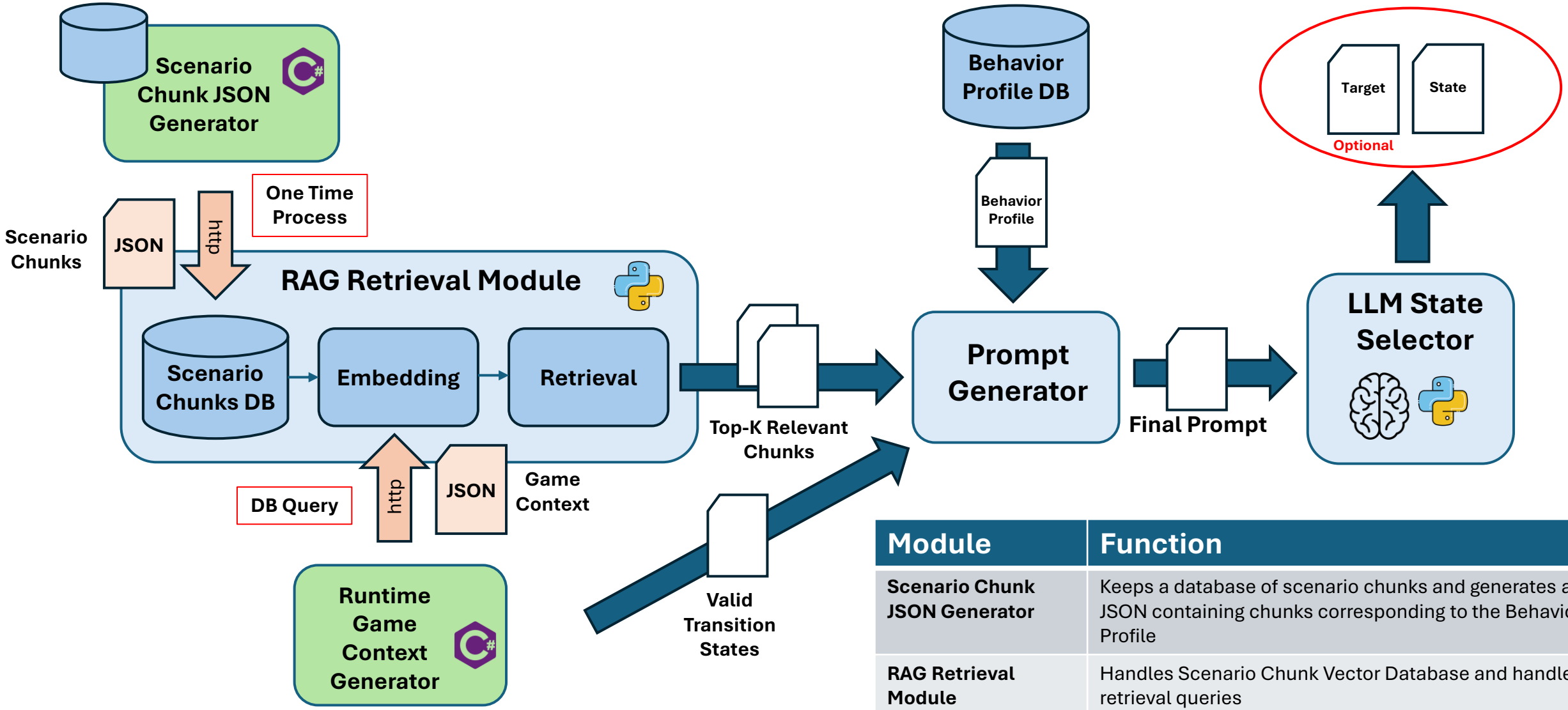
Google Colab – Codin/Testing environment
Ollama – LLM Handling
Chroma – Vector Database and Embedding Handling

Phase 3 (Planned):

Transfer/Packaging of tools for pipeline testing and developing portability to be ready for a Unity Prototype in future stage. If available connect to GPU machine provided by TIDE program. Possibly continue keeping environment in colab for T4 GPU availability.

TOOLS

VS Code – Coding IDE
Docker – Separate for game and scenario simulation C# Scripts from python database/LLM handling. Local Machine Testing.
HTTP Protocol – Messaging protocol to allow C# and Python scripts to communicate
Google Colab – Codin/Testing environment, primarily for T4 GPU use
Ollama – LLM Handling
Chroma – Vector Database and Embedding Handling
TIDE – Service allowing student access to high end machines for running tech such as LLMs (https://tide.sdsu.edu/)



High Level Architecture
Text-Based Prototype Version

Module	Function
Scenario Chunk JSON Generator	Keeps a database of scenario chunks and generates a JSON containing chunks corresponding to the Behavior Profile
RAG Retrieval Module	Handles Scenario Chunk Vector Database and handles retrieval queries
Runtime Game Context Generator	Creates test situations in the form of semi-randomly generated stats that represent a hypothetical game state
Prompt Generator	Builds final LLM prompt from retrieved scenario chunks, scenario, valid transition states, and behavior profile and then queries LLM once all aspects are available

Prompt Generation

